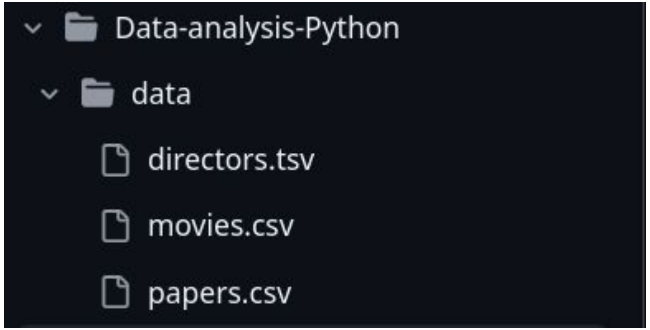
Data analysis and vizualization in Python

Data visualization in Python - RSE course

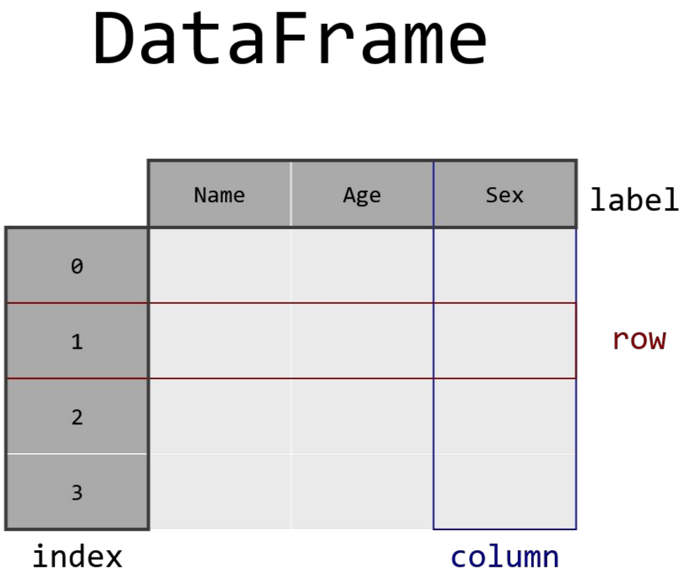
Tools and data

- libararies
  - Pandas - for statistics using tabular data
  - matplotlib - data vizualization
- datasets in this course
  - data as dictionary
  - csv - organism.csv, papers.csv
  - tsv - habitats.tsv



Pandas

- DataFrame = table like structure
- fundamental methods of dataframes - .head(), .info(), .describe()
- enables filtering, modifying and aggregating data
- loading datasets into dataframe:
    - CSV - pd.read_csv()
    - JSON - pd.read_json()
    - EXCEL - pd.read_excel()

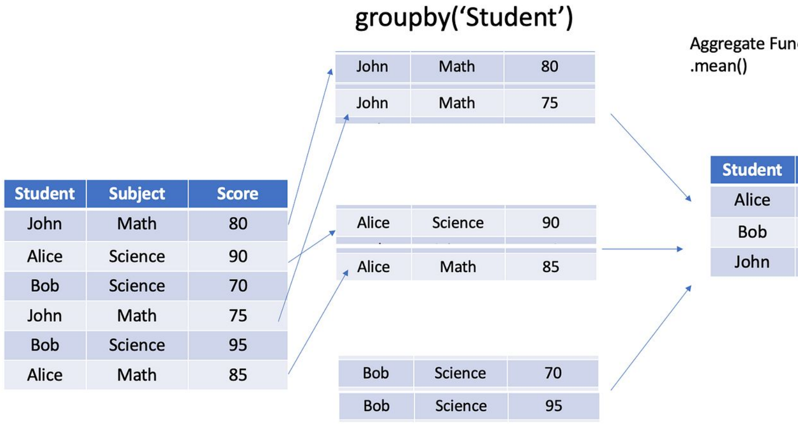


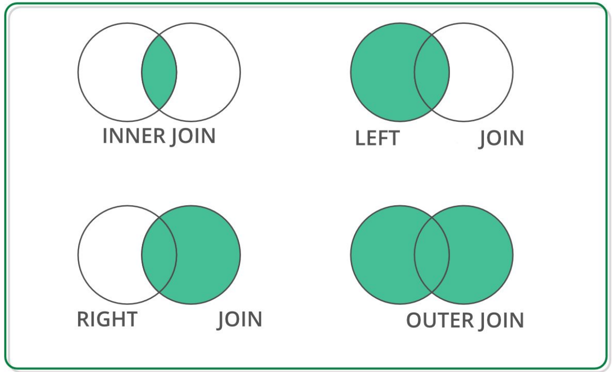
Selecting, filtering and modifying

- selecting - dataframe[“column_name”]
- filtering - conditions to filter rows where it is True
    - <, >, ==, != : dataframe[dataframe[“column_name”] > 3]
    - &, |, ~ combining conditions: dataframe[(dataframe["column_name"]=="str") | (df["column_name"]=="str1")]
- modifying
    - dataframe[“column_name”] = dataframe[“column_name”] * 10
    - dataframe.rename(columns={“column_name”: “new_column_name”})
- drop - removing specific column dataframe(“column_name”)

Aggregation

- **groupby**(“column_name”) - organising data by grouping rows from one column that share common value
- after that we can focus on another specific column and provide some info:
  - **.mean(), .count(), sum(), min(), max(), median()**
  - dataframe.groupby(“column_name”)[“column_name 2”].count()
- **merge** - two dataframes with column with same keys
  - merge(dataframe1, dataframe2, on=”name_of_column_name”, how=”left”)



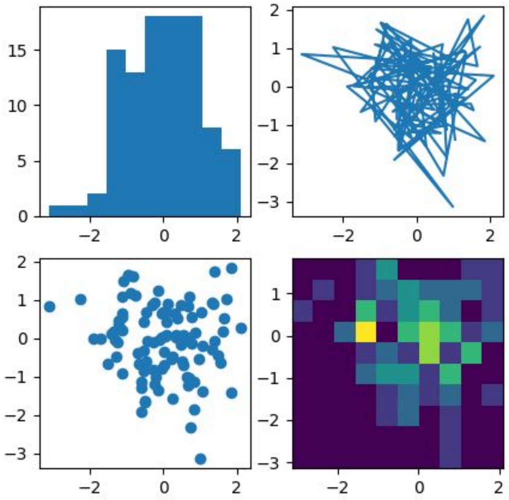


Errors in data

- pandas recognize NaN, nan, NAN, NULL, null, None, NA, N/A, ?,... and converts it to **NaN**
- how to detect issues?
  - dataframe.**info**()
  - dataframe.**describe**()
  - dataframe.**isna()** / dataframe[“column_name”].**isna()** - **finds if** there are missing data
  - dataframe.**duplicated() -** returns True for duplicate rows
- how to fix it?
  - dataframe[“column_name”].**fillna**(0) - **replaces** missing values with anything we want (e.g. 0)
    - we can also use mean - df["salary"].fillna(df["salary"].mean())
  - dataframe.**dropna()** - **removes** rows with missing values
  - dataframe[“column_name”].**astype**(int) - **converts** column to **correct type**
  - dataframe.**drop_duplicates()** - removes duplicate rows

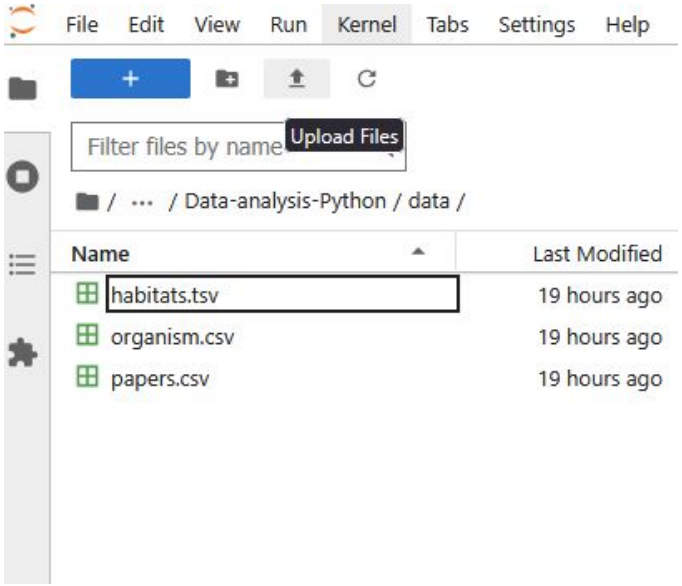
Matplotlib

- standard Python library for data visualisation
- line plots, scatter plots, histograms, boxplots, heatmaps
- plot customization - labels, titles, size, colors...
- comparison of multiple datasets



Mini project

- dataset papers.csv
- if you have some csv, tsv or excel type of dataset that is yours use it!!
  - save it into /data folder



Key takeaways

- datasets are loaded in **data frames**
- data frames can be **modified** or **filtered**
- **error data** needs to be recognized and **fixed**
- easy to compute **statistics**
- **Matplotlib** is library for **visualization**